

HDOC – 29th day

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Problem Statement:

Write a C program to perform the following menu-driven operations:

1. Count even numbers, odd numbers, positive, negative, and zero elements in an array.
2. Search for an element in an array using linear search.
3. Merge 2 arrays.
4. Exit

Algorithm:

1. Start.
2. Display the menu:
 - 1) Count array elements
 - 2) Linear search
 - 3) Merge two arrays
 - 4) Exit
3. Enter the user's choice.
4. Use switch to perform the selected operation.
5. For Choice 1, enter an array and count even, odd, positive, negative, and zero elements.
6. For Choice 2, enter an array and search for the required element using linear search.
7. For Choice 3, enter two arrays and combine them into one array.

8. For Choice 4, exit the program.
9. If the choice is invalid, display an error message.
10. Repeat the menu until Exit is selected.
11. Stop.

Test Case Table:

Test Case	Choice	Input	Expected output	Actual output	Result
1.	1	n=5, Array: 2 -3 0 4 5	Even = 3, Odd = 2, Positive = 3, Negative = 1, Zero = 1	Even = 3, Odd = 2, Positive = 3, Negative = 1, Zero = 1	Pass
2.	2	n=5, Array: 10 20 30 40 50, Search = 30	Element found at position 3	Element found at position 3	Pass
3.	2	n=5, Array: 10 20 30 40 50, Search = 60	Element not found	Element not found	Pass
4.	3	Array 1: 1 2 3, Array 2: 4 5 6	Merged Array: 1 2 3 4 5 6	Merged Array: 1 2 3 4 5 6	Pass
5.	4	_____	Exit	Exit	Pass
6.	5	_____	Invalid choice	Invalid choice	pass

kali-linux-2026.2-virtualbox-amd64 [Running] - Oracle VirtualBox

File Machine View Input Devices Help

1 2 3 4

hdoc29.c
~/Desktop/HDOC

```
35         even++;
36     else
37         odd++;
38
39     if(a[i] > 0)
40         pos++;
41     else if(a[i] < 0)
42         neg++;
43     else
44         zero++;
45 }
46
47 printf("Even = %d\n", even);
48 printf("Odd = %d\n", odd);
49 printf("Positive = %d\n", pos);
50 printf("Negative = %d\n", neg);
51 printf("Zero = %d\n", zero);
52
53 break;
54 }
55
56 case 2:
57 {
58     int key, found = 0;
59
60     printf("Enter size: ");
61     scanf("%d", &n);
62
63     printf("Enter elements: ");
64     for(i = 0; i < n; i++)
65         scanf("%d", &a[i]);
66
67     printf("Enter number to search: ");
68     scanf("%d", &key);
69
70     for(i = 0; i < n; i++)
```

C Tab W

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File Machine View Input Devices Help

1 2 3 4

hdoc29.c
~/Desktop/HDOC

Save

```
69
70     for(i = 0; i < n; i++)
71     {
72         if(a[i] == key)
73         {
74             printf("Element found at position %d\n", i + 1);
75             found = 1;
76             break;
77         }
78     }
79
80     if(found == 0)
81         printf("Element not found\n");
82
83     break;
84 }
85
86 case 3:
87 {
88     int n1, n2;
89
90     printf("Enter size of first array: ");
91     scanf("%d", &n1);
92
93     printf("Enter first array: ");
94     for(i = 0; i < n1; i++)
95     {
96         scanf("%d", &a[i]);
97         c[i] = a[i];
98     }
99
100     printf("Enter size of second array: ");
101     scanf("%d", &n2);
102
103     printf("Enter second array: ");
104     for(i = 0; i < n2; i++)
```

C Tab Width: 8 Ln 130, Col 2

Right Ctrl

kali-linux-2026.2-virtualbox-amd64 [Running] - Oracle VirtualBox

File Machine View Input Devices Help

1 2 3 4

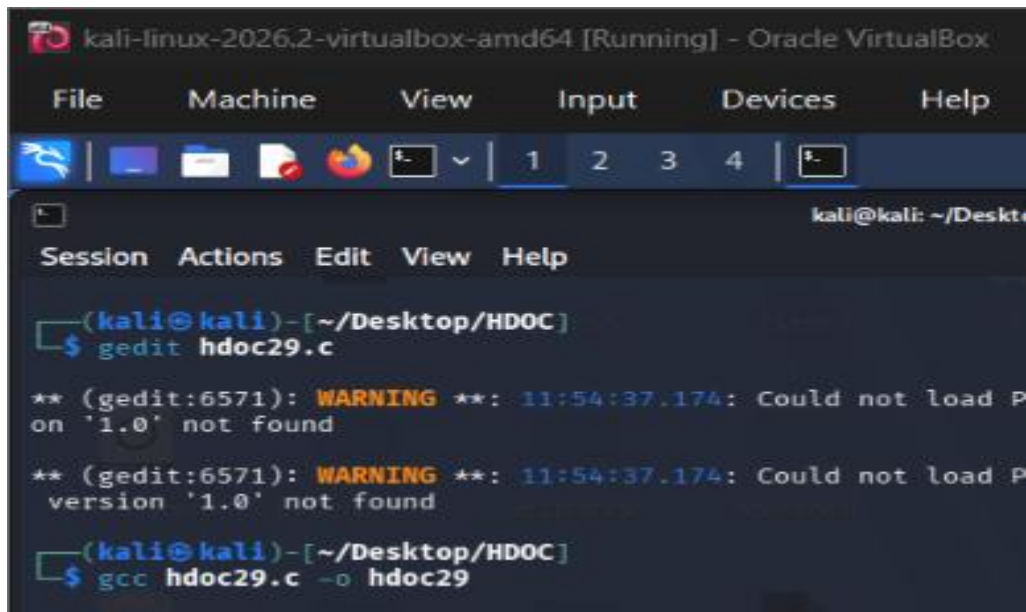
hdoc29.c
~/Desktop/HDOC Save

```
95     {
96         scanf("%d", &a[i]);
97         c[i] = a[i];
98     }
99
100    printf("Enter size of second array: ");
101    scanf("%d", &n2);
102
103    printf("Enter second array: ");
104    for(i = 0; i < n2; i++)
105    {
106        scanf("%d", &b[i]);
107        c[n1 + i] = b[i];
108    }
109
110    printf("Merged array: ");
111
112    for(i = 0; i < n1 + n2; i++)
113        printf("%d ", c[i]);
114
115    printf("\n");
116    break;
117    }
118
119    case 4:
120        printf("Exit\n");
121        break;
122
123    default:
124        printf("Invalid choice\n");
125    }
126
127    } while(choice != 4);
128
129    return 0;
130 }
```

C Tab Width: 8 Ln 1

Compile the code using:

`gcc hdoc29.c -o hdoc29`

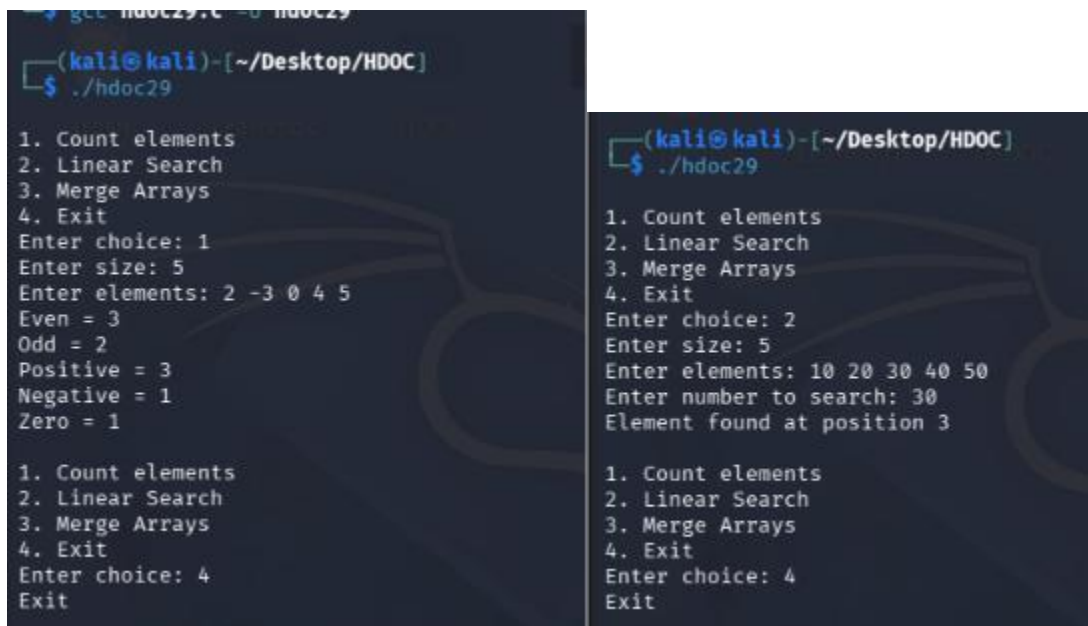


The screenshot shows a terminal window titled "kali-linux-2026.2-virtualbox-amd64 [Running] - Oracle VirtualBox". The terminal is running a Kali Linux environment. The user is in the directory `~/Desktop/HDOC`. They run `gedit hdoc29.c`, which opens the file in the gedit editor. Two warning messages are displayed: `** (gedit:6571): WARNING **: 11:54:37.174: Could not load P on '1.0' not found` and `** (gedit:6571): WARNING **: 11:54:37.174: Could not load P version '1.0' not found`. Then, the user runs `gcc hdoc29.c -o hdoc29` to compile the program.

```
kali@kali: ~/Desktop/HDOC
$ gedit hdoc29.c
** (gedit:6571): WARNING **: 11:54:37.174: Could not load P
on '1.0' not found
** (gedit:6571): WARNING **: 11:54:37.174: Could not load P
version '1.0' not found
$ gcc hdoc29.c -o hdoc29
```

Execute the code using:

`./hdoc29`



The left screenshot shows the first run of the program. The user enters choice 1, size 5, and elements 2 -3 0 4 5. The program outputs: `Even = 3`, `Odd = 2`, `Positive = 3`, `Negative = 1`, and `Zero = 1`. The right screenshot shows the second run. The user enters choice 2, size 5, and elements 10 20 30 40 50. They then enter the number to search, 30, and the program outputs: `Element found at position 3`. Both screenshots show the menu options: 1. Count elements, 2. Linear Search, 3. Merge Arrays, 4. Exit.

```
(kali@kali)~[~/Desktop/HDOC]
$ ./hdoc29
1. Count elements
2. Linear Search
3. Merge Arrays
4. Exit
Enter choice: 1
Enter size: 5
Enter elements: 2 -3 0 4 5
Even = 3
Odd = 2
Positive = 3
Negative = 1
Zero = 1
1. Count elements
2. Linear Search
3. Merge Arrays
4. Exit
Enter choice: 4
Exit

(kali@kali)~[~/Desktop/HDOC]
$ ./hdoc29
1. Count elements
2. Linear Search
3. Merge Arrays
4. Exit
Enter choice: 2
Enter size: 5
Enter elements: 10 20 30 40 50
Enter number to search: 30
Element found at position 3
1. Count elements
2. Linear Search
3. Merge Arrays
4. Exit
Enter choice: 4
Exit
```

```
(kali@kali)-[~/Desktop/HDOC]  
$ ./hdoc29
```

1. Count elements
2. Linear Search
3. Merge Arrays
4. Exit

Enter choice: 2

Enter size: 5

Enter elements: 10 20 30 40 50

Enter number to search: 60

Element not found

1. Count elements
2. Linear Search
3. Merge Arrays
4. Exit

Enter choice: 4

Exit

```
(kali@kali)-[~/Desktop/HDOC]  
$ ./hdoc29
```

1. Count Elements
2. Linear Search
3. Merge Arrays
4. Exit

Enter choice: 3

Enter size of first array: 3

Enter first array: 1 2 3

Enter size of second array: 3

Enter second array: 4 5 6

Merged array: 1 2 3 4 5 6

1. Count Elements
2. Linear Search
3. Merge Arrays
4. Exit

Enter choice: █

```
(kali@kali)-[~/Desktop/HDOC]  
$ ./hdoc29
```

1. Count Elements
2. Linear Search
3. Merge Arrays
4. Exit

Enter choice: 5

Invalid choice.

1. Count Elements
2. Linear Search
3. Merge Arrays
4. Exit

Enter choice: 4

Program ended.

```
(kali@kali)-[~/Desktop/HDOC]  
$ █
```

Program ended.

```
(kali@kali)-[~/Desktop/HDOC]  
$ ./hdoc29
```

1. Count Elements
2. Linear Search
3. Merge Arrays
4. Exit

Enter choice: 4

Program ended.